

1 · WHY THIS SMALL UNIT IS WORTH MARKS

Unit 6 is short but is examined almost entirely through "**which disease has this lesion**" objective questions. Learn each disease as a triplet: **AGENT** → **SIGNATURE LESION** → **HOW IT IS DEMONSTRATED**. Most answers turn on one named organ, one named body, or one named stain.

THE STAINS THAT APPEAR AGAIN AND AGAIN IN THIS UNIT

Warthin–Starry silver → *Clostridium piliforme* (Tyzzer's) and *Treponema* (rabbit syphilis). **Ziehl–Neelsen** → mycobacteria. **Giemsa** → blood parasites and *Clostridium piliforme*. **PAS** → fungi and glycogen.

2 · TYZZER'S DISEASE ★★★

- **Agent:** *Clostridium piliforme* (formerly *Bacillus piliformis*) — a Gram-negative, **spore-forming**, **OBLIGATE INTRACELLULAR** rod. **It cannot be grown on ordinary cell-free media** — this fact alone is a favourite question
- **Hosts:** mice, rats, **gerbils (the most susceptible)**, hamsters, rabbits, and also **foals, dogs, cats and non-human primates**. Precipitated by **stress, overcrowding, poor sanitation and corticosteroids**; spread by **spores in soiled bedding, which survive for years**

THE CLASSIC TRIAD — ALWAYS GIVE ALL THREE ORGANS ①

LIVER — multiple pinpoint to 2 mm **pale yellow-white foci of necrosis** scattered through the parenchyma (the commonest and most striking lesion)

② **INTESTINE** — **necrotising ileitis, typhlitis and colitis**; watery diarrhoea, soiled perineum, sudden death

③ **HEART** — pale streaks of **myocardial necrosis**, especially in the septum and left ventricle

MICRO — THE DIAGNOSTIC PICTURE ★ At the **margin of the necrotic focus**, inside surviving hepatocytes, lie **bundles of long, slender, filamentous bacilli arranged in a criss-cross or "haystack/faggot" pattern**. They are **poorly seen with H&E and Gram**, and are demonstrated with **WARTHIN–STARRY SILVER stain (the method of choice)**, Giemsa or PAS.

3 · PSEUDOTUBERCULOSIS & SALMONELLOSIS OF RODENTS

Disease	Agent & lesion
PSEUDOTUBERCULOSIS	Two organisms are described under this heading — know both: ① <i>Yersinia pseudotuberculosis</i> — acquired from contaminated feed, water and wild rodents/birds . Produces caseous, yellow-white "tubercle-like" NODULES in the MESENTERIC LYMPH NODES, intestinal wall, LIVER and SPLEEN , with enteritis and wasting. The nodules resemble tuberculosis but the organisms are NOT acid-fast — hence "pseudo". Zoonotic ② <i>Corynebacterium kutscheri</i> — the classic LATENT infection of rats and mice that becomes clinical only under STRESS, immunosuppression, irradiation or corticosteroids . Produces multiple caseous abscesses in the LUNGS, liver, kidneys, lymph nodes and joints . It is the standard example of an opportunistic latent laboratory-animal infection
SALMONELLOSIS	<i>Salmonella</i> Typhimurium and <i>S. Enteritidis</i> , from contaminated feed, bedding or wild rodents. Acute: septicaemia and sudden death. Subacute: catarrhal to necrotic enteritis with a distended, congested ileum and caecum , watery diarrhoea, marked SPLENOMEGALY , and multiple grey-white necrotic foci in the LIVER with mesenteric lymphadenitis; abortion and mummification in pregnant animals. An important ZOONOSIS for animal-house staff and a serious contaminant of experimental results

4 · INFECTIOUS ECTROMELIA (MOUSEPOX) ★★

- **Ectromelia virus, an ORTHOPOXVIRUS** — the mouse equivalent of smallpox. Highly contagious in a colony; spreads through skin abrasions, and also by **contaminated tumour lines and biological materials**, which is why it is a dreaded laboratory contaminant
- **The name means "missing limb"** — from the amputation of extremities in the chronic form

TWO VERY DIFFERENT FORMS — THE EXAMINER PICKS ONE

ACUTE (visceral) form — the mouse dies in a few days **WITHOUT any skin lesion**. Post-mortem shows **multiple pale foci of NECROSIS in the LIVER and SPLEEN** (splenic necrosis is very characteristic), and necrosis of Peyer's patches, lymph nodes and thymus. Mortality can approach 100% in susceptible strains.

CHRONIC (cutaneous) form — in more resistant strains. A generalised rash of **papules that ulcerate and scab**, with **oedema, necrosis and finally SLOUGHING (self-amputation) of the FEET, TAIL and SNOUT** — this is "ectromelia"; survivors are deformed and remain carriers.

Micro: epidermal **ballooning degeneration and hyperplasia** with **eosinophilic INTRACYTOPLASMIC inclusion bodies (MARCHAL bodies)** in the epithelial cells, and hepatocellular necrosis with intracytoplasmic inclusions.

5 · INFANTILE DIARRHOEA · MOUSE HEPATITIS VIRUS · LCM

★★

Disease	Agent & key point
INFANTILE DIARRHOEA (epizootic diarrhoea of infant mice, EDIM)	ROTAVIRUS. Affects suckling mice under about 2 weeks only; adults are infected but show nothing. The virus destroys the mature absorptive enterocytes at the VILLUS TIP , sparing the crypts, so the result is VILLOUS ATROPHY with a malabsorptive, osmotic diarrhoea . Clinically: yellow watery faeces, a soiled and pasted perineum ("sticky/scaly tail") , a stomach still distended with curdled milk , and runting . Morbidity is very high but mortality is low — the main loss is stunting and disrupted breeding
MOUSE HEPATITIS VIRUS (MHV)	CORONAVIRUS — probably the most prevalent and most troublesome viral contaminant of mouse colonies worldwide . Two groups of strains: Enterotropic strains — diarrhoea and death in sucklings; SYNCYTIAL (multinucleated) giant cells in the intestinal epithelium . Polytropic (respiratory) strains — spread from the nose to liver, brain and lymphoid tissue: multifocal hepatic necrosis with characteristic SYNCYTIAL GIANT CELLS , splenic and lymphoid necrosis, vasculitis, and in some strains demyelinating encephalomyelitis . In immunodeficient (nude/SCID) mice it produces a fatal chronic wasting disease . Its greatest importance is that it alters immune responses and enzyme levels and so invalidates experiments
LYMPHOCYTIC CHORIOMENINGITIS (LCM) ★	ARENAVIRUS ; the mouse is the natural reservoir and sheds virus in urine, saliva and nasal secretion for life . The classic model of how the age at infection decides the disease : ① Infected in utero or as a neonate → the mouse becomes IMMUNOLOGICALLY TOLERANT and PERSISTENTLY INFECTED with no acute illness; over months the circulating antigen–antibody complexes deposit in the kidney and cause a fatal IMMUNE-COMPLEX (Type III) GLOMERULONEPHRITIS and arteritis — "late disease" ② Infected as an adult → an immune-mediated (T-cell) ACUTE disease: lymphocytic infiltration of the MENINGES, CHOROID PLEXUS and EPENDYMA with convulsions and death. The lesion is produced by the animal's own cytotoxic T cells, not by the virus — the standard demonstration that immunopathology can be the disease ZOONOTIC — causes aseptic meningitis in people, and is a recognised hazard of handling laboratory mice, hamsters and transplanted tumour lines

6 · GUINEA PIG ★★

SCURVY — THE ONE GUINEA-PIG QUESTION YOU MUST NOT LOSE

The guinea pig, like primates, fish and fruit bats, **lacks the enzyme L-gulonolactone oxidase and therefore CANNOT SYNTHESISE VITAMIN C** — it must get ascorbic acid in the diet.

Vitamin C is needed for the **hydroxylation of proline and lysine** in collagen synthesis, so deficiency gives **defective collagen, osteoid, dentine and capillary basement membrane**.

Lesions: haemorrhages into the joints, muscles, periosteum and gums; swollen, painful joints and a reluctance to move ("scorbutic pose"); enlarged, beaded costochondral junctions; loose teeth and bleeding gums; failure of wound healing; anaemia; enlarged epiphyses with a widened, disorganised growth plate. Death from haemorrhage or secondary infection.

Condition	Key point
Cervical lymphadenitis ("LUMPS")	<i>Streptococcus equi</i> subsp. <i>zooepidemicus</i> entering through abrasions of the oral mucosa from coarse hay; produces large abscesses of the cervical (submandibular) lymph nodes that may rupture and discharge thick pus; septicæmia in some
Pneumonia	The commonest cause of death in guinea-pig colonies . <i>Bordetella bronchiseptica</i> (often carried silently by rabbits and dogs housed nearby) and <i>Streptococcus pneumoniae</i> ; cranioventral consolidation, fibrinous pleuritis, otitis media
ANTIBIOTIC-ASSOCIATED ENTEROTOXAEMIA ★	Guinea pigs have a Gram-positive-dominated caecal flora . Giving a narrow-spectrum antibiotic active against Gram-positives — penicillin, ampicillin, lincomycin, clindamycin, erythromycin — wipes it out and allows <i>Clostridium difficile</i> to overgrow. Result: fatal haemorrhagic typhlocolitis within a few days. The same principle applies to rabbits and hamsters
Pregnancy toxaemia	Two forms: a metabolic (ketotic) form in obese sows in late pregnancy — hypoglycaemia, ketonuria and a large pale fatty liver ; and an ischaemic form from compression of the uterine vessels by the gravid uterus and fat
Metastatic calcification	In older males on diets with low magnesium and a high phosphorus : calcium ratio ; gritty white deposits in the kidney, aorta, stomach, lung and skeletal muscle
Malocclusion ("slobbers")	Overgrowth of the continuously growing premolar and molar cheek teeth , which trap the tongue; drooling wets the chin and dewlap → moist dermatitis, starvation
Ectoparasites	<i>Trixacarus caviae</i> — a sarcoptid mange mite causing intense pruritus, self-trauma and even seizures ; lice <i>Gliricola porcelli</i> ; ringworm (<i>Trichophyton mentagrophytes</i>)

7 · RABBIT — BACTERIAL DISEASES ★★

Disease	Agent & lesion
PASTEURELLOSIS ("SNUFFLES")	<i>Pasteurella multocida</i> — the single most important bacterial disease of rabbits , carried subclinically in the nasopharynx and precipitated by poor ventilation, ammonia build-up and stress . It presents in several forms: ① Rhinitis "snuffles" — sneezing and a mucopurulent nasal discharge that mats the fur of the medial forepaws where the rabbit wipes its nose (a diagnostic sign) ② Otitis media/interna → "WRY NECK" (TORTICOLLIS) and loss of balance, with creamy pus in the tympanic bulla ③ Pneumonia and pleuritis — cranioventral consolidation with abscesses ④ Subcutaneous abscesses with thick, white, toothpaste-like inspissated pus (rabbit pus is characteristically caseous, not liquid) ⑤ Genital — pyometra, orchitis; and ⑥ septicaemia with sudden death
"BLUE BREASTS" (acute septic mastitis)	Chiefly <i>Staphylococcus aureus</i> , also streptococci, entering through a damaged teat in a lactating doe. The mammary glands become hot, hard, swollen and intensely painful, then progress to a dark blue-black CYANOTIC and GANGRENOUS colour — hence the name . The doe is febrile and toxæmic; the litter dies of starvation or of septicaemia acquired while nursing . Abscesses and sloughing follow in survivors
TREPONEMATOSIS (rabbit syphilis, "vent disease")	<i>Treponema paraluiscuniculi</i> — a VENEREALLY transmitted spirochaete , also passed from doe to kit at birth. It is NOT zoonotic and does NOT cause human syphilis, and it is rarely fatal — both points are often tested. Lesions: begin as erythema, oedema and small vesicles on the vulva, prepuce, scrotum and anus , which become dry, brown, crusted and scaly scabs that bleed when removed . By self-inoculation while grooming, identical lesions appear on the nose, lips and eyelids . Causes infertility and metritis. Diagnosis: SILVER (Warthin-Starry) stain or dark-field microscopy of a scraping ; the organism cannot be cultured
ENTEROTOXAEMIA	<i>Clostridium spiroforme</i> and its iota toxin . Affects mainly weanling rabbits of 4–8 weeks , precipitated by a low-fibre, high-carbohydrate diet , sudden diet change, stress or oral antibiotics . Lesions: the CAECUM is distended with dark, watery, foul-smelling fluid and gas ; the caecal and colonic mucosa shows petechial and "paint-brush" haemorrhages ; serosal haemorrhages. Death is often sudden, in the best-grown rabbits

8 · RABBIT — VIRAL AND PARASITIC DISEASES ★★★

Disease	Agent & lesion
RABBIT POX	An ORTHOPOXVIRUS closely related to vaccinia ; essentially a disease of laboratory rabbit colonies , spread by aerosol and contact. Fever, then a generalised eruption of macules and papules over the whole skin , with marked OEDEMA of the eyelids, face, lips and the anogenital region , blepharoconjunctivitis with a purulent discharge, and enlarged lymph nodes. Internally there is focal necrosis in the lung, liver, spleen and testis . Mortality is high. Intracytoplasmic inclusions
INFECTIOUS MYXOMATOSIS ★★★	MYXOMA VIRUS , a LEPORIPOXVIRUS . Transmitted MECHANICALLY by biting arthropods — mosquitoes and the rabbit flea (so outbreaks are seasonal). The natural host is the American cottontail, in which it causes only a benign fibroma; in the EUROPEAN rabbit (<i>Oryctolagus</i>) it is devastating, with mortality approaching 100% — this is the virus deliberately released in Australia to control rabbits. Gross: severe gelatinous OEDEMA of the eyelids, base of the ears, nose, muzzle and the ANOGENITAL region , producing the classic swollen, drooping " LION-HEAD " appearance; a thick purulent blepharoconjunctivitis glues the eyes shut; and multiple raised, rubbery, gelatinous cutaneous tumour-like masses called "MYXOMAS" , especially on the ears and face. Micro: the dermis is expanded by a loose, mucinous (myxomatous) matrix containing large stellate " MYXOMA CELLS ", with epidermal hyperplasia, ballooning degeneration and eosinophilic INTRACYTOPLASMIC inclusion bodies
PAPILLOMATOSIS	Two distinct viruses: ① Shope (cottontail) papillomavirus — horny,

keratinised, wart-like growths on the skin of the neck, shoulders and abdomen; historically important because these papillomas can **progress to SQUAMOUS CELL CARCINOMA**, making this the first experimental model of viral carcinogenesis in a mammal.

② **Rabbit oral papillomavirus** — small, grey-white, sessile **nodules on the UNDERSIDE OF THE TONGUE and floor of the mouth** of young rabbits; benign and self-limiting

COCCIDIOSIS ★

The commonest parasitic disease of rabbits.

Two forms — always answer both:

① **HEPATIC coccidiosis** — *Eimeria stiedae*, which is unique in parasitising the **EPITHELIUM OF THE BILE DUCTS**. The **liver is enlarged and studded with sharply outlined white to yellow nodules and cords that follow the course of the bile ducts**; the ducts are thick-walled and full of creamy material packed with oocysts. Micro: **marked bile-duct (biliary) epithelial HYPERPLASIA with papillary projections, and all stages of the parasite inside the epithelial cells**. Icterus, ascites, and death in young rabbits.

② **INTESTINAL coccidiosis** — *E. magna*, *E. irresidua*, *E. perforans*, *E. flavescens* — catarrhal to haemorrhagic enteritis, diarrhoea, dehydration and poor growth

MITE INFESTATION

Psoroptes cuniculi — "**EAR CANKER**", by far the commonest. Thick, dry, greyish-brown, crumbly, flaky **crusts filling the ear canal and pinna**; intense irritation with head shaking and ear scratching; may extend to **otitis media/interna and torticollis**, or become secondarily infected. *Cheyletiella parasitovorax* — "**WALKING DANDRUFF**", large surface-living mites causing scurf and mild alopecia over the shoulders and back; **ZOONOTIC**, giving an itchy rash on handlers.

Sarcoptes scabiei and *Notoedres* — burrowing mange of nose, lips and feet with thick crusts and intense pruritus

9 · VIRAL DISEASES OF WILD ANIMALS ★★

Disease	Agent & pathology in wildlife
WEST NILE FEVER	FLAVIVIRUS , transmitted by <i>Culex</i> mosquitoes. Wild BIRDS are the amplifying/reservoir hosts; corvids (crows, jays, magpies) die in large numbers and a die-off of crows is the classic early warning of an outbreak. Horses and humans are DEAD-END hosts (they do not develop enough viraemia to infect a mosquito). Lesions: in birds — myocardial necrosis, splenomegaly, encephalitis, haemorrhage. In horses — non-suppurative POLIOENCEPHALOMYELITIS concentrated in the GREY MATTER of the BRAIN STEM and SPINAL CORD , with perivascular cuffing, gliosis, neuronal necrosis and microglial nodules; clinically ataxia, muscle fasciculation and recumbency. Zoonotic
RABIES	Same Lyssavirus , but in wildlife the epidemiology differs: the reservoirs in India include the jackal, fox, mongoose and stray dog , and elsewhere bats, raccoons and skunks . Wild carnivores show the furious form — loss of fear of man, daytime activity, aggression, aimless wandering — and herbivores such as deer show the dumb/paralytic form . Same pathology: non-suppurative polioencephalitis with NEGRI BODIES ; diagnosis by FAT on brain . Rabies in a wild animal that has bitten people is a public-health emergency
FOOT AND MOUTH DISEASE	Affects wild cloven-footed species — gaur, deer (chital, sambar), blackbuck, nilgai, wild boar and Asian elephants . Same lesions: vesicles and erosions on the tongue, dental pad, interdental cleft and coronary band , lameness, and "TIGER HEART" myocarditis in the young , which causes the deaths. Outbreaks in national parks usually spill over from infected livestock grazing at the forest fringe — an important interface problem in India
POX	Camelpox in camels; buffalopox (zoonotic, teat and udder lesions); MONKEYPOX in non-human primates and rodents, a serious zoonosis ; elephant pox; and capripox in wild ruminants. All show the same sequence macule → papule → vesicle → pustule → scab with ballooning degeneration and INTRACYTOPLASMIC inclusion bodies
KYASANUR FOREST DISEASE ★★ ("monkey fever")	A FLAVIVIRUS first recognised in 1957 in the Kyasanur forest of Shimoga district, KARNATAKA, INDIA — so it is a favourite Indian question. Transmitted by the hard tick Haemaphysalis spinigera ; small forest rodents and shrews maintain it, and monkeys — the black-faced langur and the red-faced bonnet macaque — die in large numbers . A monkey die-off in the forest precedes human cases , so it is used as a sentinel. Lesions: a haemorrhagic fever — widespread haemorrhages in the gut, lungs and serosae, focal necrosis of the liver and kidney , interstitial pneumonia, and erythrophagocytosis in the spleen, liver and lymph nodes . In people it causes a severe haemorrhagic fever; forest workers and veterinarians are at risk
INFECTIOUS HEPATITIS VIRUS	Canine adenovirus-1 in wild carnivores — foxes, wolves, jackals, coyotes and bears . In addition to the usual centrilobular hepatic necrosis with Cowdry type A intranuclear inclusions and gall-bladder wall oedema, it characteristically produces "FOX ENCEPHALITIS" — vasculitis of CNS vessels with haemorrhage and non-suppurative encephalitis , giving convulsions and paralysis. Related canine adenovirus-2 causes kennel cough

10 · BACTERIAL DISEASES OF WILD ANIMALS ★★

Disease	Pathology in wildlife
ANTHRAX	The classic cause of mass die-offs of wild herbivores — grazing animals ingest spores concentrated at drying waterholes and on disturbed soil, especially at the end of a dry season. Vultures, hyaenas and flies then spread the spores widely . Same rules as in livestock and they are always examined: a wild herbivore found dead with dark tarry unclotted blood from the natural orifices, absent or incomplete rigor mortis and a huge soft "blackberry-jam" spleen must NOT be opened in the field . Take a peripheral blood smear and stain by polychrome methylene blue for McFADYEAN'S reaction . Dispose by incineration or deep burial with quicklime ; the carcass site remains infective for decades. Carnivores are relatively resistant but may get the pharyngeal/intestinal form
TUBERCULOSIS	<i>Mycobacterium bovis</i> in deer, wild boar, bison and antelope — often maintained in a wildlife reservoir that reinfects cattle. Captive ELEPHANTS are important because they usually get M. tuberculosis FROM PEOPLE — a REVERSE zoonosis , presenting as chronic weight loss with a trunk-wash-positive culture. Lesions are the usual caseous granulomas with epithelioid and Langhans giant cells in lung and lymph nodes , but calcification is often minimal in deer, in which the lesions may be soft and abscess-like . Confirmed by Ziehl-Neelsen and culture
COLIBACILLOSIS	<i>E. coli</i> in hand-reared and captive-born neonates, almost always secondary to FAILURE OF PASSIVE TRANSFER when the young is separated from the dam or fed an unsuitable milk replacer. Produces watery diarrhoea and dehydration , and in the septicaemic form omphalitis, polyarthritis, meningitis and fibrinous polyserositis . A leading cause of death in zoo nurseries
CLOSTRIDIAL INFECTIONS	Enterotoxaemia (<i>Cl. perfringens</i>) in captive and hand-reared ungulates after sudden access to rich concentrate — pulpy kidney, haemorrhagic enteritis. Blackleg (<i>Cl. chauvoei</i>) in wild bovids — dark crepitant muscle. Tetanus after wounds, snare injuries and darting. BOTULISM (<i>Cl. botulinum</i> type C) is the great one in wildlife — it causes massive die-offs of WATERFOWL at shallow, warm, drying wetlands, where the birds eat maggots from rotting carcasses; they show flaccid paralysis of the neck ("LIMBERNECK") , wings and legs and drown , and there are no gross lesions

11 • PARASITIC DISEASES OF WILD ANIMALS

Disease	Key point in wildlife
TRYPANOSOMOSIS	<i>Trypanosoma evansi</i> (surra) is the important one in India — it affects camels most severely , and also deer, tigers, leopards and elephants in captivity , where infected buffalo or camels nearby act as the source and <i>Tabanus</i> flies transmit it MECHANICALLY . Big cats often acquire it by eating infected meat . Signs: intermittent fever, progressive anaemia and cachexia , ventral oedema, dullness, and terminal nervous signs . Post-mortem: splenomegaly, lymphadenopathy, serous atrophy of fat, pale carcass, hydropericardium and non-suppurative meningoencephalitis . Diagnosis: wet blood film for motile organisms, Giemsa-stained smear, mouse inoculation, PCR
BABESIOSIS	<i>Babesia</i> species in wild bovids, deer and wild canids, transmitted by ixodid ticks . Free-living wildlife is usually in endemic stability and shows no disease, acting as a reservoir. Clinical disease appears in stressed, translocated, captured or immunosuppressed animals — fever, intravascular haemolysis with HAEMOGLOBINURIA , jaundice, anaemia and splenomegaly. Diagnosis: pear-shaped paired piroplasms inside the red cells on a Giemsa-stained smear
THEILERIOSIS	<i>Theileria</i> species in wild bovids and deer, transmitted by <i>Hyalomma</i> and <i>Rhipicephalus</i> ticks . Marked lymph node enlargement , fever, anaemia and jaundice, "punched-out" abomasal ulcers , and pulmonary oedema. Diagnosis by KOCH'S BLUE BODIES (schizonts) in lymphocytes of a lymph-node aspirate. As with babesiosis, wildlife often carries the parasite silently and infects introduced exotic or crossbred stock

12 • NUTRITIONAL DEFICIENCY DISEASES OF WILD ANIMALS

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Almost all of these are diseases of **CAPTIVITY** — a free-ranging animal selects a balanced diet, a caged or hand-reared one eats what it is given.

Deficiency	Disease produced
Calcium / phosphorus / vitamin D (NUTRITIONAL SECONDARY HYPERPARATHYROIDISM) ★	The commonest and most important nutritional disease of captive wildlife . Caused by an ALL-MEAT diet in hand-reared carnivores (muscle meat is very low in calcium and high in phosphorus, giving a badly inverted Ca : P ratio), and by lack of sunlight in indoor enclosures. Low ionised calcium → sustained parathyroid hormone release → bone is resorbed and replaced by fibrous tissue = FIBROUS OSTEODYSTROPHY . Lesions: soft, thin, poorly mineralised bones with pathological folding fractures, bowed limbs, a rubbery mandible, collapsed vertebrae with paralysis , reluctance to move, and tetany. Classic in hand-reared tiger and leopard cubs, primates and captive reptiles . Called rickets if the growth plates are still open, osteomalacia in adults

Vitamin A

Common in captive carnivores, primates and reptiles on poor diets: **night blindness and xerophthalmia, squamous metaplasia and keratinisation of glandular epithelium**, respiratory infection, poor growth, and in reptiles the classic **swollen closed eyelids from squamous metaplasia of the harderian gland**. Conversely, **hypervitaminosis A** from an all-liver diet causes **deforming cervical spondylosis** in captive cats

Vitamin E / selenium

WHITE MUSCLE DISEASE / nutritional muscular dystrophy in captive ungulates, especially **deer, antelope and zoo-reared calves** on stored feed: **bilaterally symmetrical chalky white streaks in skeletal muscle and heart**, Zenker's degeneration, ↑ CK and AST. Also **steatitis ("yellow fat disease")** in fish-eating animals and captive felids

THIAMINE (B₁) ★

Occurs in animals fed **large amounts of RAW FISH, which contains THIAMINASE** — captive **fish-eating birds, otters, seals, dolphins, crocodiles and fur animals**. This is **"CHASTEK PARALYSIS"**: anorexia, ataxia, head tremor, **opisthotonus/star-gazing** and convulsions, and in ruminants **POLIOENCEPHALOMALACIA** with laminar cortical necrosis that **autofluoresces under UV light**. Prevented by cooking the fish or supplementing thiamine

Vitamin C

Scurvy in captive **primates, guinea pigs, fruit bats and some fish**, which cannot synthesise ascorbic acid — haemorrhages, painful swollen joints, gum disease and failure of wound healing

Iodine • copper • cobalt • iron

Goitre in captive ungulates and reptiles on goitrogenic diets; **copper deficiency** → ataxia, achromotrichia and anaemia in wild ruminants on deficient range; **iron deficiency** anaemia in hand-reared young

Protein–energy malnutrition

Free-ranging wildlife in drought or overstocked habitat: **emaciation, SEROUS ATROPHY OF FAT in the coronary groove, perirenal region and bone marrow** (the marrow becomes gelatinous and watery — the standard field indicator of starvation), hypoproteinaemic oedema and heavy parasite burdens

NOT NUTRITIONAL, BUT THE WILDLIFE LESION YOU MUST KNOW

★ **CAPTURE MYOPATHY (exertional rhabdomyolysis)** — follows the **chase, capture, restraint or transport** of a wild animal, especially deer, antelope, blackbuck and zebra. Extreme exertion with catecholamine release causes **lactic acidosis and Zenker's (coagulative) necrosis of skeletal and cardiac muscle: pale, dry, chalky-white streaked muscle, myoglobinuria with dark brown urine and renal tubular necrosis**, and death hours to days later. **Prevention** — **minimise chase time, use proper chemical immobilisation, avoid the heat of the day**.

13 · NECROPSY OF LABORATORY AND WILD ANIMALS — THE PRACTICAL RULES

- **Legal position:** a wild animal necropsy needs the **authority of the Forest Department** under the Wild Life (Protection) Act; it is usually done **in the presence of forest officials**, and in a suspected poaching or poisoning case it is treated exactly like a **veterolegal post-mortem** — photographs, numbered samples, seals and a documented chain of custody
- **Always exclude ANTHRAX first** in any herbivore found dead — blood smear before the knife
- **Look for human interference:** snare and wire marks, bullet or pellet wounds (radiograph if possible), missing claws, teeth, horn, skin or gall bladder, and **evidence of deliberate poisoning of a carcass** — carnivore deaths in India are commonly due to **pesticide-laced baits**, so collect **stomach content, liver and kidney frozen, with NO preservative**, for toxicology
- **Decomposition is usually advanced** in field cases; interpret cautiously and separate post-mortem change from lesion
- **Laboratory animals:** the value of the necropsy is often to detect **subclinical colony infections that invalidate experiments** (MHV, LCM, *Corynebacterium kutscheri*). Perform under sterile conditions; sample for serology and PCR as well as histopathology, and remember that **LCM, salmonellosis, and Cheyletiella are ZONOOSES**

14 · BLANK-FILL DRILL — UNIT 6

- Tyzzer's disease is caused by Clostridium piliforme, which is an obligate intracellular organism
- Tyzzer's disease affects the liver, intestine and heart
- The bacilli of Tyzzer's disease lie in a criss-cross pattern and are shown by the Warthin-Starry silver stain
- Gerbils are the most susceptible rodent to Tyzzer's disease
- Pseudotuberculosis nodules are not acid-fast, unlike tuberculosis
- Corynebacterium kutscheri causes a latent infection that flares under stress
- Salmonellosis in rodents causes splenomegaly with hepatic necrotic foci
- Infectious ectromelia is caused by an orthopoxvirus and the name means missing limb
- The acute form of mousepox causes necrosis of the liver and spleen with no skin lesions
- Mousepox inclusions are called Marchal bodies and are intracytoplasmic
- Epizootic diarrhoea of infant mice is caused by rotavirus and gives villous atrophy
- Mouse hepatitis virus is a coronavirus and produces syncytial giant cells
- Lymphocytic choriomeningitis virus is an arenavirus; the natural reservoir is the mouse
- Mice infected with LCM in utero become immunologically tolerant and develop immune-complex glomerulonephritis
- In adult mice the LCM lesion is caused by the animal's own cytotoxic T cells
- Guinea pigs cannot make vitamin C because they lack L-gulonolactone oxidase

- Scurvy causes haemorrhage into the joints and gums and beading of the costochondral junctions
- "Lumps" in guinea pigs is cervical lymphadenitis caused by Streptococcus zooepidemicus
- Penicillin given to a guinea pig causes fatal enterotoxaemia due to Clostridium difficile
- Snuffles in rabbits is caused by Pasteurella multocida; middle-ear spread gives torticollis
- Rabbit pus is characteristically thick and caseous, not liquid
- Blue breasts of rabbits is acute mastitis, usually caused by Staphylococcus aureus
- Rabbit syphilis is caused by Treponema paraluis-cuniculi and is not zoonotic
- Rabbit enterotoxaemia is caused by Clostridium spiroforme and its iota toxin
- Myxomatosis is caused by a leporipoxvirus spread mechanically by mosquitoes and fleas
- Myxomatosis produces the lion-head appearance and stellate myxoma cells in a mucinous matrix
- Shope papilloma of rabbits may progress to squamous cell carcinoma
- Hepatic coccidiosis of rabbits is caused by Eimeria stiedae, which parasitises the bile duct epithelium
- Ear canker in rabbits is caused by Psoroptes cuniculi
- Cheyletiella parasitovorax causes walking dandruff and is zoonotic
- West Nile virus is a flavivirus spread by Culex mosquitoes; birds are the amplifying hosts
- Horses and humans are dead-end hosts for West Nile virus
- Kyasanur forest disease was first recognised in Karnataka and is spread by Haemaphysalis spinigera
- A die-off of monkeys precedes human cases of Kyasanur forest disease
- Canine adenovirus-1 in foxes causes fox encephalitis with CNS vasculitis
- Captive elephants usually acquire Mycobacterium tuberculosis from humans — a reverse zoonosis
- Botulism causes mass mortality of waterfowl and produces limberneck
- Surra in captive tigers is often acquired by eating infected meat
- The commonest nutritional disease of hand-reared carnivores is nutritional secondary hyperparathyroidism from an all-meat diet
- Feeding raw fish causes thiamine deficiency because fish contains thiaminase
- Chastek paralysis is thiamine deficiency in fur-bearing and fish-eating animals
- Gelatinous watery bone marrow indicates serous atrophy of fat, that is starvation
- Capture myopathy produces Zenker's necrosis of muscle and myoglobinuria
- Wildlife necropsy requires the authority of the Forest Department and toxicology samples must carry no preservative